INCH-POUND

MIL-B-83831
12 JANUARY 1994

MILITARY SPECIFICATION

BOOTH, CLEANING, WATER BASE SOLVENT SPRAY (FOR ELECTRONIC MEASURING INSTRUMENTS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of floor mounted, manually operated, nonrecirculating water base solvent spray booth for cleaning of electronic measuring instruments.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

ZZ-H-521	Hose and Hose Assemblies, Nonmetallic Spray
GGG-G-770	Gun, Air Blow (Blast Cleaning)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technology & Industrial Services Division, SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6425 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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MILITARY

MIL-P-116	Preservation, Methods Of
MIL-C-87936	Cleaning Compounds, Aircraft Exterior Surfaces, Water Dilutable

STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standards for Federal Services
FED-STD-595	Color

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Block and Waterproofing; With Appropriate Test Methods
MIL-STD-1190	Minimum Guidelines For Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Department of the Navy, Defense Printing Service, Detachment Office, 700 Robbins Ave., Bldg 4D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D3951 Standard Practice For Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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AMERICAN WELDING SOCIETY, Inc.

AWS D1.1 Structural Welding Code Steel

AWS D1.2 Structural Welding Code Aluminum

(Application for copies of AWS publications should be addressed to the American Welding Society, Inc., 550 N. W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 National Electrical Code

(Application for copies of NFPA publications should be addressed to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. This specification provides for first article testing (see paragraphs 4.2 and 4.3).

3.2 Description. The booth shall consist essentially of a booth enclosure, booth enclosure stand, cleaning solution/rinse water spray gun, air blast gun, necessary hoses, controls, lighting, exhaust fan and connections for required utilities. Spray pressure is furnished by means of shop air.

3.3 Function. The booth, when operated as intended, shall clean electronic instruments, especially those normally used in Air Force precision measurement laboratories, by spraying with a water base cleaning solution followed by a clear water spray rinse followed by moisture removal by means of an air blast gun. The booth shall accommodate items, for cleaning, having dimensions 24 inches by 24 inches by 24 inches and weighing 150 pounds. The booth operator shall stand outside the booth with arms and hands extending through the open side of the booth enclosure to manually direct spray and air against the workpiece on the turntable within the booth enclosure and to manually rotate turntable so as to reposition the workpiece. Over spray collected in booth enclosure basin shall not be recirculated, but rather shall drain to the waste water connection.

3.3.1 Cleaning solution. The booth shall be suitable for use with 6:1 to 3:1 mixture of tap water to MIL-C-87936, type I, alkaline, water base cleaning compound as the spraying cleaning solution.

3.4 Details of components.

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3.4.1 Booth enclosure. The booth enclosure shall be a sheet steel covered rectangular structure open at the front. The booth enclosure shall be watertight. The interior shall be smooth to prevent pocketing and to facilitate cleaning. The bottom shall be formed into a basin or sump. The low point of the basin shall be provided with a drain connected to the waste drain nipple. An exhaust air opening shall be provided in the upper rear of the enclosure. Sheet steel comprising cabinet enclosure side and top surfaces and the basin shall be not less than Manufacturer's Standard (AISI) Gage No. 16. The booth enclosure shall be fitted with two hooks for convenient storage of the spray and air dry guns. Booth enclosure front opening shall not be less than 29 inches wide by 28 inches high. Booth enclosure interior front to back dimensions shall not be less than 30 inches. The booth enclosure shall be suitable for bench mounting or stand mounting.

3.4.2 Stand. The booth shall be furnished with a separate structural steel stand or legs for support of the booth enclosure. Interconnection between the stand or legs and the booth enclosure shall be secure. The stands or legs vertical dimension shall be such that the turntable top surface shall be between 39 and 40 inches above floor level. The stand or legs shall be gusseted and braced for rigidity. Stand legs shall be fitted with feet drilled for floor anchor bolts. Stand or legs shall be fitted with a shelf, 12 inches high minimum, for storage of cleaning concentrate containers.

3.4.3 Turntable. A manually rotatable horizontal turntable for support and positioning of workpieces shall be located within the booth enclosure. Diameter of the turntable shall be not less than 24 inches. The turntable shall have a fitted rubber mat or shall be coated with not less than 0.015 inches of black fused nylon. The turntable shall rotate without wobble when supporting a 150-pound load uniformly distributed over an 18-inch square placed at any location on the turntable. Turntable bearing(s) shall be suitable for use in a water laden atmosphere.

3.4.4 Lighting. Booth enclosure interior lighting shall be provided. Lighting shall consist of two 150 watt sealed reflector, incandescent lamps, each in an individually adjustable lamp holder. Lamp holders shall be so located that workpiece illumination is shadowless and positional adjustment is convenient and exposure to spray is minimized.

3.4.5 Exhauster. An electric motor driven fan/blower shall be furnished that shall convey all water vapor and fumes out of the booth enclosure. The exhauster shall be mounted on/in the top rear panel of the booth enclosure. The exhauster shall be easily removed and accessible for cleaning and replacement. Exhauster shall be sized to produce an average face velocity in the booth enclosure front opening of not less than 125 ft/min when exhausting against 1/2" water static pressure.

3.4.6 Spray gun. A cleaning solution and rinse water spray gun shall be provided with the booth. Spray gun shall be intended for one hand use, shall be well balanced and comfortable to hold and convenient to use. Spray gun shall be fitted with a manual trigger. Spray gun shall be fitted with a hanger eye for

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convenient storage on the booth enclosure hooks. Spray gun shall have a manually variable orifice with knurled locking ring or nut to control the degree of fineness/coarseness of the delivered spray. Spray gun shall incorporate a nozzle extension such that the spray tip is between 10 and 12 inches from the hand position. Spray gun shall be served by an air hose and a cleaning solution/rinse water hose. Action of the reduced shop air shall withdraw solution from the reservoir, convey solution to the spray gun and deliver solution in the form of a spray to the workpiece. Means shall be furnished to limit both cleaning solution and rinse water delivered by the spray gun to between 1/4 and 1/8 gpm when the reduced air pressure is 50 psig and when water service pressure is between 40 and 60 psig. Spray gun shall withstand multiple four foot drops onto a concrete floor without permanent deformation that affects serviceability.

3.4.7 Air dry gun. An air dry gun shall be provided with the booth. The air dry gun shall be any 1/4 inch NPT Style A (with hanger) gun, except type III, conforming to Federal Specification GGG-G-770 Gun, Air Blow (Blast Cleaning).

3.4.8 Hoses. Air, water and solution hose assemblies shall be furnished as necessary. Hoses serving cleaning spray gun and air blast gun shall be of sufficient length to insure freedom of movement of the guns. Hoses serving cleaning spray gun and air blast gun shall be lightweight and flexible consistent with service requirements. The material hose assembly shall withstand a burst test pressure of not less than four times the maximum pressure developed by a stall or bypass pressure. The outer hose covering shall be abrasion and oil resistant. The inside material of the hose shall be resistant to aromatic solvents. Each end of the material hose shall have a swivel, female coupling and a 60 degree included bevel taper seat. The air hose assembly shall conform to ZZ-H-521. The air hose assembly shall withstand a 200 psig working pressure and a burst pressure of at least 800 psig. The air supply hose shall have a quickly detachable air coupler with an automatic shut valve at one end.

3.4.9 Cleaning solution reservoir. A cleaning solution reservoir having a nominal capacity of not less than one gallon shall be furnished. Reservoir shall be of a translucent material such that the solution level within the reservoir can be observed by the booth operator. The reservoir shall have a top opening for convenient manual addition of cleaning compound to the reservoir. Filling of reservoir with tap water shall be accomplished by activation of fill control (see paragraph 3.4.10.b). Water fill line to the reservoir shall be fitted with a device to limit filling flow to 2 $\pm$1/4 gallons per minute when water pressure varies from 40 to 60 psig. Turbulence from automatic filling of the reservoir with tap water shall mix the cleaning compound with the incoming water. Aspirating action of shop air at the cleaning gun shall withdraw cleaning solution from the reservoir for delivery of cleaning solution to the cleaning gun.

3.4.10 Controls. Controls shall be mounted outside the booth enclosure and shall be readily accessible to the operator. Controls shall be prominently and permanently marked as to function. Controls shall include:

a. Line switch that shall energize lighting, exhauster and control circuit.

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b. Fill control that shall control filling of solution reservoir with tap water.

c. Clean/rinse control which shall determine whether spray gun delivers cleaning solution or rinsing tap water.

d. Pressure regulator that shall regulate 200 psig supply air down to between 20 and 80 psig. The regulator shall be adjustable for any pressure required for operation within the 20-80 psig range. The regulator shall hold the set pressure without creep until changed by the operator.

e. Dial indicating pressure gauge that shall display reduced air pressure.

3.5 Dimensions. There are no restrictions on the booth overall dimensions.

3.6 Weight. Dry weight of the booth shall not exceed 300 pounds.

3.7 Utilities. The booth shall operate as intended when connected to the following utilities:

a. Electric power - 120 volts, 1 phase, 60 Hz.

b. Tap water - 60 psig.

c. Shop compressed air - 200 psig.

d. Waste drain.

e. Exhaust air venting.

3.8 Utility connections. Provisions for connection of all utilities, required by paragraph 3.7 Utilities, shall be provided at the rear of the booth enclosure as follows:

a. Junction box for electric power.

b. Three-fourths inch IPS nipple for tap water supply.

c. Three-eighths inch IPS nipple for shop compressed air supply.

d. Three-fourths inch IPS nipple for waste drain.

e. Breeching (duct flange) for exhaust air venting.

3.9 General. Booth shall be furnished complete, except for connection of utilities and cleaning compound, so that it can be used and operated as intended. The booth and its components shall be suitable for their intended use. The booth shall be rugged and sturdy and shall withstand the strains and stresses, impacts and vibration incident to shipping, storage, base transport and service. The

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booth shall be free of any defect that affects serviceability and appearance. Air and liquid connections shall not leak. The booth shall permit ease of access for installation, servicing and maintenance. All parts subject to wear, breakage and distortion shall be adjusted, repaired or replaced by use of common hand tools. Screw threads shall be in the inch system and shall be in accordance with the applicable requirements of FED-STD-H28.

3.10 Safety standard. Electrical wiring and associated electrical components shall conform to and be installed in accordance with National Fire Protection Association (NFPA) Standard No. 70.

3.11 Materials. Materials not specifically designated herein shall be of good commercial quality. Wood shall not be used as a booth material. Contact between dissimilar metals, especially brass, copper or steel in contact with either aluminum or magnesium, shall be avoided. Materials shall be either inherently corrosion resistant or shall be suitably protected against corrosion in accordance with MIL-STD-889.

3.12 Construction. Welding shall be sound, smooth and free from pits, holes and fissures. Sheet metal shall be free from kinks and waviness. Sheet metal surfaces shall not "oil can." All sheet metal edges shall be smooth. Bolt holes shall be accurately punched and reamed or drilled, and shall have all burrs removed. Washers and lock washers shall be provided in accordance with commercial practice. All bolts, nuts and screws shall be tight and shall be installed to prevent loss of tightness. All wiring shall be enclosed in metal tubing where possible; and in all cases, all wiring shall be secured and free from scuffing, kinking or straining.

3.13 Workmanship. Workmanship shall be of good commercial quality and shall be such as to produce a booth of a neat general appearance and capable of functioning and serving as intended. The booth, including all parts and assemblies shall be fabricated, assembled and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; thoroughness of welding; alignment of parts and tightness of assembly screws and bolts; uniformity and appearance of painting. Welding shall be in accordance with AWS D1.1 or D1.2, as applicable.

3.14 Finish. The booth and stand or legs shall be painted in accordance with the manufacturer's standard commercial practice. The finish shall be designed to prevent corrosion. Color shall be gray in accordance with FED-STD-595.

3.15 Marking.

3.15.1 Identification. The booth shall be identified by means of an identification plate permanently secured to one exterior side surface of the booth enclosure; the identification plate shall be permanently and legibly marked with the following information:

a. Booth, cleaning, water base solvent (for electronic instruments).

b. Assigned NSN.

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c. Contract number.

d. Manufacturer's name or trademark.

e. Manufacturer's part number.

f. Manufacturer's serial number. (Optional)

g. U.S. Government Property

3.15.2 Instruction. Instruction plates shall be provided as needed for proper operation and service of the unit.

3.16 Interchangeability. All booths furnished under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies and spare parts. The manufacturer's drawing numbers and part numbers shall be the same. All parts shall be manufactured to definite standard, clearances and tolerances in order that identically numbered parts can be functionally and dimensionally interchanged without requiring modification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification. The inspection and testing of the booth shall be classified as follows:

a. First article(see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article.

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4.3.1 First article test sample. One booth shall be subjected to the inspection and tests specified in 4.5 Test method. The sample shall be identical in all respects to the booths to be produced as production items.

4.3.2 First article test report. Upon completion of the first article inspection and tests, a test report shall be prepared in accordance with MIL-STD-831 and the following shall be incorporated as a part of the report:

a. Manufacturer's assembly drawing of the booth.

b. Eight by ten inch black and white photographs of the booth.

c. Checklists used during examination (see 4.5.1) and operation (see 4.5.2) testing.

4.4 Quality conformance inspection. Inspection of each production booth shall consist of examination (see 4.5.1) and operation (see 4.5.2).

4.5 Test methods.

4.5.1 Examination. The booth, including its components, shall be inspected to determine compliance with the requirements herein with respect to design, configuration, materials, dimensions, weight, construction, workmanship, marking and finish. Noncompliance shall cause rejection. First article examination shall be accomplished by means of a contractor developed checklist which shall list booth characteristics subject to examination, method of verification and results obtained. This same checklist shall be used during acceptance examination except that weight of booth need not be verified.

4.5.2 Operation. The booth shall be connected to the necessary utilities and shall be operated to determine proper functioning of guns, controls, lights and exhauster. All connections shall be inspected for air and liquid leaks. Preproduction operation testing shall be accomplished following a contractor developed checklist which shall list booth operating features subject to test, method of test and results obtained. This same checklist shall be used to conduct acceptance operations testing.

5. PACKAGING

5.1 General requirements. The booth and components shall be clean and dry in accordance with applicable procedures, that are non-injurious to the parts, of MIL-P-116. Preservatives (P type) and methods (IC-1, IIa) shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.1 Disassembly. The booth may be disassembled to effect savings in cubage, protect parts vulnerable to damage and to assure all surfaces requiring a preservative are accessible for application. The item shall not be disassembled to the extent that it would require special skills or tools to reassemble.

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5.1.2 Matchmarking. Removed parts and mating parts remaining on the equipment shall be match marked to facilitate reassembly.

5.1.3 Exterior surfaces. Unpainted, unfinished, ferrous metal surfaces of bolts, nuts, screws, washers, brackets, fittings, stands, and other exterior surfaces requiring a contact preservative shall be coated with type P-2 preservative and protected with grease proof barrier material. Unfinished ferrous metal information places shall be coated with varnish.

5.2 Preservation. Preservation shall be Level A, C or commercial, as specified.

5.2.1 Level A.

5.2.1.1 Booth, stand and turntable. These items shall be provided unit protection in accordance with Method III of MIL-P-116. Wrap and cushioning shall be provided to prevent marring or denting.

5.2.1.2 Light fixtures. The light fixtures shall be provided cushioning to preclude any damage.

5.2.1.3 Electric motor. All openings to electric windings shall be sealed with pressure sensitive tape to prevent entry of any foreign material. A tag shall be affixed with the precautionary statement "REMOVE PACKAGING TAPE PRIOR TO OPERATION OF THE MACHING."

5.2.1.4 Switches. Switches shall be protected with a minimum of 1 inch non-dusting, non-corrosive cushioning material and taped in place.

5.2.1.5 Spray and air guns. Spray and air guns shall be provided preservation, as necessary, to assure any ferrous metal parts will not corrode and protected by either boxing or cushioning the guns from being damaged or causing damage to other items.

5.2.1.6 Hose and hose fittings. Open ends of all hose or hose fittings shall be sealed with plastic caps or plugs or with waterproof, pressure sensitive tape that will provide protection against entry of any foreign material. Lengths of hose shall be coiled to a safe minimum diameter. The coils shall be uniform, compact and of a diameter to prevent deformation or kinking. Each coiled hose shall be secured approximately equidistant a minimum of three places with cotton twine or pressure sensitive tape. Loose fittings shall be placed in a transparent plastic bag; sealed and secured on the inside of the coil.

5.2.1.7 Reservoir. Openings in the reservoir shall be closed with weatherproof, pressure sensitive tape and larger openings shall be covered with barrier material taped in place. If reservoir is ferrous metal, it shall be preserved with type P-3 preservative before closing.

5.2.1.8 Controls. The controls, mounted on the outside of the booth, shall be provided with a fitted piece of hardboard or fiberboard and secured in place with pressure sensitive tape.

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5.2.1.9 Technical publications. The technical publications for each booth shall be preserved in accordance with Method IC-1 or IC-3. Special note: When more than one technical publication is required for a booth, the publications shall be preserved as a set in one Method IC-1 or IC-3 bag.

5.2.1.10 Consolidation. Preserved components and technical publications shall be placed either in the booth and cushioned to restrict movement and not damage the booth or, if the booth is broken down, the components shall be placed in a snug-fitting fiberboard box.

5.3 Packing. Packing shall be Level A, B, C or Commercial; as specified.

5.3.1 Level A. The booth with consolidated components shall be packed, as applicable to its weight, in a wood box. Wood boxes shall conform to PPP-B-601, overseas type, style optional or PPP-B-621, Class 2, style optional. The closure and strapping shall be as specified in the applicable specification. Blocking, bracing, cushioning and anchoring shall be in accordance with MIL-STD-1186.

5.3.2 Level B. Level B packing shall be the same as Level A, except that domestic or class 1 type wood boxes shall be used in lieu of overseas or class 2.

5.3.3 Level C. Level C packing shall be in accordance with the requirements of MIL-STD-1190.

5.3.4 Commercial. Commercial packing shall be in accordance with the requirements of ASTM D3951.

5.4 Marking. All marking, whether military or commercial, shall be in accordance with the requirements of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The booth is intended for cleaning of precision electronic measuring instruments by removal of conductive contamination by means of a water base cleaning solution spray, water rinse spray and air blast water removal. Final drying of equipment so cleaned is to be effected by use of Oven, Infrared drying (low temperature), PD WRAMA/MMEME 4430-434, NSN 4430-00-472-3342.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification.

b. Preservation, packaging and packing required (see 5.1).

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6.3 Subject term (key word) listing.

Enclosure

Air dry gun

Regulator

Spray Gun

Exhauster

Turntable

Custodian:
Air Force - 99

Reviewer:
Army - AL

Preparing activity:
Air Force - 82

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